

	Business Intelligence	Data Analytics	Data visualization
Purpose	BI focuses on transforming raw data into meaningful and useful information for business analysis purposes. It aims to support better business decision-making.	Data analytics involves examining, cleaning, transforming, and modeling data to discover useful information, draw conclusions, and support decision-making.	Data visualization involves presenting data in graphical or pictorial format to help people understand complex data sets and see patterns, trends, and correlations.
Components	1. Data Sources: - Internal Data: Data from within the organization, such as CRM systems, ERP systems, and financial databases. - External Data: Data from external sources, such as market research, social media, and third-party data providers. 2. Data Integration: - ETL (Extract, Transform, Load): The process of extracting data from various sources, transforming it into a suitable format, and loading it into a data warehouse. - Data Pipelines: Automated workflows that move and transform data from source systems to the data warehouse. 3. Data Warehousing: - Data Warehouse: A centralized repository that stores integrated data from multiple sources, optimized for querying and analysis. - Data Marts: Subsets of the data warehouse tailored to specific business lines or departments.	1. Data Collection: - Data Sources: Identify and gather data from various sources such as databases, CRM systems, social media, sensors, or external data providers. - Data Ingestion: Importing and storing the collected data into a centralized system or data warehouse. 2. Data Preparation: - Data Cleaning: Removing or correcting inaccuracies, inconsistencies, and duplicates. - Data Transformation: Converting data into a suitable format for analysis, including aggregating, normalizing, and encoding data. - Data Integration: Combining data from different sources to create a unified dataset. 3. Data Storage: - Data Warehousing: Organizing and storing large volumes of data in a data warehouse for easy access and analysis. - Data Lakes: Storing raw or unprocessed data that can be transformed later as needed. 4. Data Exploration and Analysis: - Descriptive Analytics: Summarizing and describing historical data to understand what happened. - Diagnostic Analytics: Investigating data to determine why certain events or outcomes occurred. - Predictive Analytics: Using statistical models and machine learning algorithms to forecast future trends or outcomes.	1. Data Collection & Preparation: - Data Sources: Gather data from databases, APIs, spreadsheets, or other data repositories. - Data Cleaning: Remove errors, fill in missing values, and ensure data consistency. - Data Transformation: Convert raw data into a suitable format for visualization. 2. Visualization Tools: Tools and software that enable the creation of visual representations of data. - BI Platforms: Power BI, Tableau, Looker - Data Visualization Libraries: D3.js, Highcharts, Plotly - Programming Languages: Python (Matplotlib, Seaborn), R (ggplot2) 3. Types of Visualizations: Different types of visual representations to display various data insights. - Charts and Graphs: Bar charts, line charts, pie charts, scatter plots

	4. Data Modeling: Conceptual, Logical, and Physical Models: Models that define the structure, relationships, and storage of data in the data warehouse. Star Schema and Snowflake Schema: Common data modeling techniques used to organize and optimize data for querying. 5. Data Mining: Pattern Recognition: Identifying patterns and trends within large data sets using statistical and computational methods. Predictive Analytics: Using historical data to forecast future trends and outcomes. 6. Online Analytical Processing (OLAP): OLAP Cubes: Multidimensional data structures that allow for complex queries and analysis. Drill-Down and Roll-Up: Techniques to navigate through different levels of data granularity. 7. Reporting and Querying: Standard Reports: Predefined reports that provide regular insights on key metrics. Ad-Hoc Queries: Custom queries created by users to explore specific data points. 8. Dashboards and Visualizations:	Prescriptive Analytics: Recommending actions based on predictive insights to optimize decision-making. 5. Data Visualization: Charts and Graphs: Bar charts, line charts, scatter plots, etc., to visualize data patterns and trends. Dashboards: Interactive visual interfaces that display key metrics and performance indicators. Geospatial Maps: Visual representation of data in relation to geographical locations. Infographics: Combined visuals and data points in a single image to convey complex information. 6. Advanced Analytics Techniques: Machine Learning: Algorithms that learn from data to make predictions or identify patterns. Natural Language Processing (NLP): Analyzing and interpreting human language data. Data Mining: Discovering hidden patterns and relationships within large datasets using statistical and computational methods. Text Analytics: Extracting meaningful information from text data. 7. Data Interpretation and Communication: Reporting: Generating regular and ad-hoc reports to share analytical findings. Storytelling: Presenting data insights in a compelling narrative to engage stakeholders. Actionable Insights: Translating data analysis results into actionable recommendations for decision-makers. 8. Data Governance: Data Quality Management: Ensuring the accuracy, completeness, and reliability of data. Compliance: Adhering to data privacy laws and regulations like GDPR, HIPAA, etc. Security: Protecting data from breaches and unauthorized access.	Maps: Geospatial visualizations like heatmaps, choropleth maps Dashboards: Interactive displays combining multiple visualizations Infographics: Visual summaries combining text, images, and charts 4. Design Elements: Key elements used in visualizing data effectively. Color: Use of colors to differentiate data points and highlight key information. Shapes: Different shapes can represent various categories or data types. Size: Adjusting the size of visual elements to convey quantitative data. Text and Labels: Adding titles, labels, and annotations to clarify data points. 5. Interactivity: Features that allow users to interact with the data visualization. Filters and Drill-downs: Enable users to explore data by filtering or drilling down into specific details. Tooltips: Show additional information when hovering over data points. Zoom and Pan: Allow users to zoom in and out or pan across large data sets. 6. Dashboarding:
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	• Interactive Dashboards: Visual interfaces that display real-time data and key performance indicators (KPIs). • Visualization Tools: Software like Power BI, Tableau, and Looker that create visual representations of data.	9. Tools and Technologies: • Software: Tools like Python, R, SAS, Apache Spark, SQL, and specialized analytics platforms. • Platforms	Creating comprehensive and interactive dashboards that provide an overview of key metrics and insights. • Layout and Design: Arranging visual elements logically. • Widgets: Interactive components such as dropdowns, sliders, and checkboxes. • Real-time Data Integration: Connecting to live data sources to display up-to-date information.
	9. Business Performance Management: • KPIs and Metrics: Key performance indicators and metrics used to measure business success. • Scorecards and Performance Monitoring: Tools to track and report on business performance.		7. Storytelling: Presenting data in a narrative format to highlight insights and tell a story. • Context: Providing background information to explain the data. • Flow: Structuring the visualizations in a logical sequence. • Annotation: Adding notes and explanations to guide the audience through the insights.
	10. Data Governance and Security: • Data Quality Management: Ensuring the accuracy, completeness, and reliability of data. • Data Security: Protecting data from unauthorized access and breaches. • Compliance: Adhering to legal and regulatory requirements like GDPR, HIPAA, etc.		8. Best Practices: Guidelines to ensure effective and accurate data visualization. • Simplicity: Avoid clutter and focus on key messages. • Accuracy: Ensure the visual representation accurately reflects the data. • Consistency: Use consistent colors, fonts, and styles. • Accessibility: Make visualizations easily understandable for all audiences.
	11. Metadata Management: • Metadata Repository: A database that stores information about data sources, definitions, and relationships. • Data Dictionary: Documentation that provides details about data elements and their meanings.		
	12. User Training and Support: • Training Programs: Educating users on how to use BI tools and interpret data.		

	• Support Services: Providing assistance and troubleshooting for BI users. 13. Collaboration and Sharing: • Collaborative Platforms: Tools that allow users to share reports, dashboards, and insights. • Version Control: Ensuring that users access the most up-to-date and consistent information.		9. Publishing & Sharing: Distributing visualizations to stakeholders. • Web Embedding: Integrating visualizations into websites or web applications. • Reports: Including visualizations in detailed reports. • Collaboration Platforms: Sharing via platforms like Slack, SharePoint, or email. 10. Evaluation & Iteration: Assessing the effectiveness of visualizations and making improvements. • Feedback: Collecting feedback from users to identify areas for enhancement. • A/B Testing: Comparing different visualization designs to determine the most effective one. • Continuous Improvement: Updating visualizations as new data becomes available or business needs change.
Tools	Power BI, Tableau, Looker, BusinessObjects.	R, Python, SAS, Apache Spark, Google Analytics.	Tableau, Power BI, D3.js, Matplotlib, Adobe Illustrator.
Users	BI Analysts, Managers, Decision Makers.	Data Analysts, Data Scientists, Marketing Analysts.	Data Journalists, BI Analysts, UX Designers, Executives.
Types	1. Traditional BI • Descriptive Analytics: Focuses on summarizing historical data to understand what happened. It involves reporting, querying, and trend analysis. • Key Tools: SQL-based databases, OLAP cubes, data warehouses.	1. Descriptive Analytics • Purpose: To summarize and describe historical data. • Focus: What happened? • Examples: Monthly sales reports, website traffic summaries, and revenue growth trends.	1. Bar Charts and Column Charts • Use: Compare quantities across categories. • Example: Sales performance by region or product.

2. Operational BI

- **Real-Time Analytics:** Provides near-instant access to data to support day-to-day operations. It focuses on monitoring current performance and activities.
- **Use Cases:** Call center performance dashboards, real-time stock level monitoring.
- **Key Tools:** Streaming data platforms, in-memory databases.

3. Self-Service BI

- **Empowerment:** Allows non-technical users to create their own reports and visualizations without needing IT support.
- **Key Features:** Ad-hoc reporting, drag-and-drop interfaces, and user-friendly dashboards.
- **Key Tools:** Power BI, Tableau, Qlik Sense.

4. Advanced & Predictive Analytics

- **Predictive Modeling:** Uses historical data to predict future outcomes. It involves statistical models and machine learning algorithms.
- **Use Cases:** Sales forecasts, customer churn predictions.
- **Key Tools:** R, Python, SAS, Apache Spark.

5. Embedded BI

- **Integration:** Embeds BI capabilities directly into business applications to provide context-specific analytics.

2. Diagnostic Analytics

- **Purpose:** To understand why something happened.
- **Focus:** Why did it happen?
- **Examples:** Analyzing the cause of a drop in sales, investigating customer churn reasons, and finding the root cause of product defects.

3. Predictive Analytics

- **Purpose:** To forecast future outcomes based on historical data.
- **Focus:** What is likely to happen?
- **Examples:** Sales forecasting, predicting customer behavior, and identifying potential equipment failures.

4. Prescriptive Analytics

- **Purpose:** To provide recommendations for action based on predictive analytics.
- **Focus:** What should we do?
- **Examples:** Optimizing supply chains, suggesting marketing strategies, and recommending medical treatments.

Additional Analytics Features

Inferential Analytics

- **Purpose:** To make inferences about a population based on sample data.
- **Focus:** Making predictions or generalizations.
- **Examples:** Market research surveys, opinion polls, and A/B testing results.

Exploratory Data Analysis (EDA)

- **Purpose:** To discover patterns, spot anomalies, and test hypotheses.
- **Focus:** What are the data's underlying patterns?

2. Line Charts

- **Use:** Display trends over time.
- **Example:** Monthly revenue growth, stock prices.

3. Pie Charts and Donut Charts

- **Use:** Show parts of a whole.
- **Example:** Market share distribution, budget allocation.

4. Scatter Plots

- **Use:** Show relationships between two variables.
- **Example:** Correlation between advertising spend and sales.

5. Histograms

- **Use:** Display frequency distribution of a dataset.
- **Example:** Distribution of test scores, age groups.

6. Heatmaps

- **Use:** Display data intensity or frequency using color.
- **Example:** Website click maps, correlation matrices.

7. Treemaps

- **Use:** Show hierarchical data as nested rectangles.

	• Use Cases: In-app analytics, embedded dashboards within CRM systems. • Key Tools: API-based integration, custom BI widgets.	• Examples: Initial data analysis steps using visualizations and summary statistics to understand data distribution and relationships.	• Example: Portfolio allocation, organizational structure.
	6. Cloud BI • Scalability: Utilizes cloud services to deliver BI capabilities, offering flexibility and on-demand scalability. • Key Features: SaaS BI platforms, cloud-based data storage and processing. • Key Tools: AWS QuickSight, Google Data Studio, Microsoft Azure Synapse.	Causal Analytics • Purpose: To identify cause-and-effect relationships. • Focus: How does changing one factor affect another? • Examples: Understanding the impact of marketing campaigns on sales, and the effect of new policies on employee productivity.	8. Bubble Charts • Use: Show three variables with x, y coordinates and bubble size. • Example: Revenue (x), profit (y), and market size (bubble size).
	7. Mobile BI • Accessibility: Provides BI insights on mobile devices, enabling on-the-go access to data. • Use Cases: Mobile dashboards, alerts, and notifications		9. Area Charts • Use: Show cumulative totals over time. • Example: Accumulated revenue over the years.
			10. Box Plots (Box-and-Whisker Plots) • Use: Display data distribution summaries. • Example: Test scores distribution, income ranges.
			11. Geospatial Maps • Use: Visualize geographical data. • Example: Sales by state, population density maps.
			12. Gantt Charts • Use: Display project schedules. • Example: Project timelines, task durations.

			13. Waterfall Charts • Use: Show cumulative effect of sequential positive and negative values. • Example: Financial statements, profit analysis. 14. Network Diagrams • Use: Visualize relationships between nodes. • Example: Social network connections, IT network topology. 15. Word Clouds • Use: Display text data frequency. • Example: Keywords in survey responses, website tags. 16. Infographics • Use: Combine data visuals and narrative elements. • Example: Marketing materials, educational content.
Techniques	Business Intelligence (BI) techniques are essential for transforming data into actionable insights that empower organizations to make informed decisions. Here are some key BI techniques:	Data analysis involves examining, cleaning, transforming, and interpreting datasets to discover useful information, draw conclusions, and support decision-making. Here are some essential data analysis techniques:	Data visualization techniques are crucial for making data more understandable and actionable. Here are some popular techniques used in data visualization: 1. Bar Chart • Purpose: Compare quantities across categories.

	1. Data Warehousing • Purpose: Collect, store, and manage large volumes of data from multiple sources. • Components: ETL (Extract, Transform, Load), OLAP (Online Analytical Processing) cubes, data marts. • Techniques: Data warehousing design, data modeling (star schema, snowflake schema), and data integration. 2. Data Mining • Purpose: Discover patterns and relationships within large datasets. • Components: Association rule learning, clustering, classification, regression. • Techniques: Market basket analysis, anomaly detection, trend analysis. 3. Reporting and Querying • Purpose: Generate reports and queries to summarize and analyze data. • Components: Standard reports, ad-hoc queries, drill-down reports. • Techniques: SQL querying, report generation, data aggregation. 4. Dashboarding and Visualization • Purpose: Present data visually to help stakeholders quickly understand insights. • Components: Interactive dashboards, charts, graphs, maps.	1. Descriptive Analysis • Purpose: Summarizes and describes the main features of a dataset. • Techniques: Measures of central tendency (mean, median, mode), measures of dispersion (variance, standard deviation), frequency distributions, and data visualization (charts, graphs). 2. Inferential Analysis • Purpose: Makes inferences about a population based on a sample. • Techniques: Hypothesis testing, confidence intervals, regression analysis, ANOVA (Analysis of Variance). 3. Exploratory Data Analysis (EDA) • Purpose: Identifies patterns, detects anomalies, tests hypotheses, and checks assumptions with the help of summary statistics and graphical representations. • Techniques: Data visualization (histograms, box plots, scatter plots), summary statistics, correlation matrices. 4. Predictive Analysis • Purpose: Uses historical data to predict future outcomes. • Techniques: Regression models (linear regression, logistic regression), time series analysis, machine learning algorithms (decision trees, random forests, neural networks). 5. Prescriptive Analysis	• When to Use: For comparing discrete data or display the frequency of categorical data. • Example: Sales performance by region. 2. Line Chart • Purpose: Show trends over time. • When to Use: For illustrating data points connected by straight lines to show changes in value. • Example: Stock prices over a year. 3. Pie Chart • Purpose: Display parts of a whole. • When to Use: For showing the proportion of categories within a whole dataset. • Example: Market share distribution. 4. Scatter Plot • Purpose: Show relationships between variables. • When to Use: For identifying correlations between two variables. • Example: Relationship between advertising spend and sales revenue. 5. Histogram • Purpose: Display frequency distribution. • When to Use: For representing the distribution of a dataset. • Example: Age distribution of a population.
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	• Techniques: Data visualization best practices, storytelling with data, real-time dashboards. 5. Performance Management • Purpose: Monitor and manage business performance using key metrics. • Components: Key Performance Indicators (KPIs), scorecards, performance dashboards. • Techniques: Balanced scorecard, KPI tracking, performance benchmarking. 6. Predictive Analytics • Purpose: Forecast future trends and outcomes based on historical data. • Components: Statistical modeling, machine learning algorithms, time series analysis. • Techniques: Predictive modeling, regression analysis, decision trees. 7. Prescriptive Analytics • Purpose: Recommend actions based on predictive insights. • Components: Optimization models, simulation, scenario analysis. • Techniques: Prescriptive modeling, Monte Carlo simulation, sensitivity analysis. 8. Data Governance • Purpose: Ensure data quality, security, and compliance within the organization.	• Purpose: Provides recommendations for actions based on predictive analysis. • Techniques: Optimization models, simulation, scenario analysis, decision analysis. 6. Diagnostic Analysis • Purpose: Determines why something happened by examining data. • Techniques: Root cause analysis, drill-down analysis, correlation analysis, and cohort analysis. 7. Text Analysis (Text Mining) • Purpose: Extracts valuable insights from text data. • Techniques: Natural Language Processing (NLP), sentiment analysis, topic modeling, keyword extraction. 8. Cluster Analysis • Purpose: Groups a set of objects in such a way that objects in the same group (cluster) are more similar to each other than to those in other groups. • Techniques: K-means clustering, hierarchical clustering, DBSCAN (Density-Based Spatial Clustering of Applications with Noise). 9. Factor Analysis • Purpose: Identifies underlying relationships between variables by grouping them into factors. • Techniques: Principal Component Analysis (PCA), Exploratory Factor Analysis (EFA), Confirmatory Factor Analysis (CFA).	6. Heatmap • Purpose: Represent data intensity using color. • When to Use: For visualizing matrix data and showing the magnitude of values. • Example: Website click maps. 7. Treemap • Purpose: Show hierarchical data. • When to Use: For visual segmentation and comparison of hierarchical datasets. • Example: Portfolio allocation. 8. Bubble Chart • Purpose: Show three variables. • When to Use: For visualizing data with two quantitative variables and a third variable represented by bubble size. • Example: GDP (x), population (y), and country size (bubble). 9. Gantt Chart • Purpose: Display project timelines. • When to Use: For managing and scheduling project tasks. • Example: Project management plans. 10. Waterfall Chart • Purpose: Display cumulative value changes.
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	• Components: Data quality management, data security policies, compliance monitoring. • Techniques: Data cleansing, data auditing, data stewardship.		
	9. Data Integration • Purpose: Combine data from various sources to provide a unified view. • Components: Data blending, ETL processes, data migration. • Techniques: Data mapping, data consolidation, real-time data integration.		
	10. Self-Service BI • Purpose: Empower non-technical users to perform their own data analysis. • Components: User-friendly BI tools, interactive dashboards, drag-and-drop interfaces. • Techniques: Ad-hoc analysis, data exploration, user training and support.		
	11. Benchmarking and Competitive Analysis • Purpose: Compare business performance against industry standards and competitors. • Components: Benchmark reports, competitive intelligence, market analysis. • Techniques: SWOT analysis, benchmarking studies, competitive profiling.	10. Time Series Analysis • Purpose: Analyzes data points collected or recorded at specific time intervals to identify trends, cycles, and seasonal patterns. • Techniques: Moving averages, exponential smoothing, ARIMA (AutoRegressive Integrated Moving Average) models.	• When to Use: For understanding the sequential impact of transactions on a given value. • Example: Financial statements analysis.
		11. Monte Carlo Simulation • Purpose: Uses randomness to solve problems that might be deterministic in nature. • Techniques: Generating random samples, probability distributions, and simulating outcomes.	11. Violin Plot • Purpose: Show the distribution and density of a dataset. • When to Use: For comparing the distribution of multiple groups. • Example: Distribution of test scores between classes.
		12. Anomaly Detection • Purpose: Identifies unexpected items or events in datasets that differ from the norm. • Techniques: Z-score, DBSCAN, Isolation Forest, and time-series-based anomaly detection.	12. Box Plot • Purpose: Show data distribution summaries. • When to Use: For displaying the spread and skewness of data. • Example: Examining a dataset's quartiles.
			13. Area Chart • Purpose: Show cumulative totals. • When to Use: For visualizing trends over time with stacked data. • Example: Accumulated revenue over the years.
			14. Geospatial Map • Purpose: Visualize geographical data. • When to Use: For connecting data and geographical locations. • Example: Population density maps.

	12. Advanced Analytics • Purpose: Perform complex analysis to gain deeper insights. • Components: Machine learning, AI-driven analytics, big data processing. • Techniques: Deep learning, natural language processing (NLP), big data analytics. 13. Collaboration and Sharing • Purpose: Facilitate the sharing of insights and collaboration among stakeholders. • Components: BI portals, shared dashboards, report distribution. • Techniques: Collaborative BI, report sharing, data annotation.		15. Chord Diagram • Purpose: Visualize relationships between categories. • When to Use: For representing interrelationships in complex datasets. • Example: Trade relationships between countries. 16. Sankey Diagram • Purpose: Show flow and volume. • When to Use: For visualizing flow from one set of values to another. • Example: Energy distribution flows. 17. Word Clouds • Purpose: Display text data frequency. • When to Use: For visualizing the importance or frequency of words. • Example
Summary	BI is a broader discipline that includes data analytics and visualization as part of its process.	Data Analytics is more focused on finding insights, conducting statistical analyses, and making predictions based on data.	Data Visualization serves as a critical element for both BI and data analytics, helping to make data understandable and actionable through graphical representations.